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Stock Recommendation System for Better Investment Plan

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***Abstract*— In the volatile landscape of financial markets, making well-informed decisions for intraday trading and long-term investment necessitates an integrated analysis of real- time stock data and the contextual information provided by news articles. This paper presents an innovative stock recommendation system that leverages advanced Machine Learning (ML) techniques and Natural Language Processing (NLP) to deliver timely, data-driven recommendations by assessing both risk and profit potential. Our system employs ML algorithms to process live stock price, trading volume, and technical indicators, enabling real-time assessment of a stock's performance. Developing a stock recommendation system is a challenging undertaking that frequently calls for a multidisciplinary team with experience in data science, machine learning, and finance. Simultaneously, it harnesses NLP models to extract sentiment and contextual information from news articles, providing insights into market sentiment and potential impact on stock behavior. By merging these two streams of data, our system offers a comprehensive view of stock's dynamics.**

***Index Terms*—Stock Prediction, Machine Learning, Investment Decisions, Stock Returns, Recommendation, LSTM**

I. Introduction

Stock prediction is regarded as one of the long existing research problem. One significant goal is to create efficient models for prediction that can accurately project if future Stock prices will increase or decrease. The major and challenging issue that needs to be rapidly resolved in the finance business is the accurate prediction of stock trends, which is the foundation for investors to gain significant profits. A stock price forecast is important because it is both business people and regular folks use. Participants in stock market activity will either gain or lose money. It is an unreliable system. Because a variety of factors might affect price changes, factors like news, statistics from social media, fundamentals, historical prices, governmental bonds, business output, and it is difficult to create an acceptable model for national economics. There are established numerous factors for modeling the pattern flow of stock performance employee and public feedback, as well as specific events. One of the key difficulties of the complexity of stock prediction is that stock price volatility is frequently unpredictable since the stock market is the consequence of multi-element and multi-vector synergy. Investors in the stock market may feel overwhelmed and intimidated by their options when faced with such a wide variety of stocks.

Stock prediction is a difficult task that attracts the interest of scholars from a variety of professions. The research articles that we have analyzed in current domain [1] revealed that the combinations of CNN with Word2Vec and CNN with BERT were the most successful in predicting stock movements over the next day, week, and month. While both CNN+Word2Vec and CNN+BERT demonstrated similar prediction performance, BERT's feature extraction process was computationally more complex compared to Word2Vec. The results indicated that the choice of news platform could affect the performance of prediction models. The number of financial terms and sentiment scores conveyed in news articles emerged as crucial factors for constructing effective stock prediction models. In a study listed as in [2], it was discovered that while the attention weights of the headlines and bodies views of most news samples are relatively low, there are some samples where the attention weight of these two views is still greater than that of sentiments. This indicates that sentiment is not always the most crucial element of news views. A smart combination of headline, body and sentiment is necessary. Although using a single news sentiment as the input of the model can be effective, it is important to note that the Boson NLP tool is not perfect in reproducing news sentiment and may misinterpret it.

The rest of this paper is structured as follows. Section II gives Detailed literature review and Section III an overview of the Research Motivation and insights derived from various research articles. Section IV describes the proposed methodology including Objective of this project, Model Architecture, Dataset Identification, Feature Selection , Data Preprocessing. Section V delivers the experimental results and finally Section VI concludes the paper with mentioning all the references in Section VII. We concluded by summarizing the article and outlining the path it should take.

II. Literature Review

In this section , we have discussed about recent research articles on the theme of Stock prediction and recommendation. A stock recommendation system based on Twitter analysis computes a sentiment index for all companies. This index is then used to predict which companies are good to invest in, based on both real stock prices and Twitter data. The companies are recommended based on their specific classification group.

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The study also looked for any correlation between market conditions and share values, as well as the parameters that influence these aspects. The researchers focused on analyzing the features of the customer community, such as age, employment level, and

income ranges, in buying and selling stocks.

Recommendation system built based on trust [4] demonstrated that mutual fund transactions with an augmented assistance of social network analysis can be utilized to build a trustworthy stock market portfolio recommendation system. They show that trust based stock recommendation system is an effective approach to find trustworthy stocks based on credible mutual fund investment patterns giving high returns.

The model presented in [5] mentioned that three different approaches that can be used to build a personalized stock recommender system to help users in their decision-making when investing in stocks. In the framework of monitoring learning processes, several findings were recorded regarding the estimation of stock market forecasts, and suggestions were made.

This model is specially designed [6] employing multiple regression analysis, a statistical method used to examine the relationship between a dependent variable (stock market trends in this case) and multiple independent variables (which could include economic indicators, financial metrics, or other relevant factors). The authors collect and analyze data from the stock market and various sources to build their predictive mode. The research paper presents the results of the multiple regression analysis and demonstrates how different independent variables can influence and predict stock market trends. It likely discusses the statistical significance of these variables and their coefficients in the regression model.

By extracting tweets from StockTwits over pipeline API and treated them for Natural Language Processing (NLP) and sentiment analysis. SVM is applied to forecast the sentiment of each tweet. The primary objective of this research is to investigate whether sentiment analysis of StockTwits, a social media platform where users discuss and share information about stocks and financial markets, can enhance the accuracy of predicting stock price movements. After predicting sentiment they mined past data from Yahoo Finance. They established a model for stock market prediction by means of stock price data and sentiment score to guess the change in stock market. The paper concludes by citing Whether integrating StockTwits sentiment analysis enhances stock price prediction. [7]

Improved news sentiment analysis [8] likely describes the methodology used, which may include the following steps: Collecting a dataset of news articles for sentiment analysis. Applying deep learning models, such as neural networks or recurrent neural networks (RNNs), to capture and analyze sentiment in news text. Experimenting with various deep learning architectures and techniques to identify the most effective approach for news sentiment analysis. Experimenting with different deep learning architectures and methods to find the best method for sentiment analysis of news.

Two different approaches [9] likely describes the methodology used, which may include the following: Collecting a dataset of news articles related to financial markets and stocks. The primary objective of this research is to investigate and compare the effectiveness of various machine learning and deep learning methods in predicting stock prices using news articles as a data source. Applying different machine learning and deep learning algorithms, such as decision trees, support vector machines, recurrent neural networks (RNNs), and other relevant models. Evaluating and comparing the performance of these models in predicting stock price movements based on the sentiment and information contained in the news articles.

Different text classification algorithms [10] likely covers the following aspects: A detailed review of various text classification algorithms, which could include traditional machine learning techniques, deep learning methods, and hybrid models. Discussion of the key principles and components of these algorithms, such as feature extraction, model training, and classification. Evaluation criteria for assessing the performance of text classification algorithms.

Financial firms also struggle to suggest equities that fit the trading styles of their customers. The majority of consumers and the market revolve around online trading, which is open to anyone and allows members to quickly participate from anywhere and gain even with no stock market experience. However, the reality is very different. The opportunity here is that if we could provide the clients some better projections, they would get more profit from our observations and advice. In the investor's decision-making process, emotion is important.

III. Research Motivation

Most researchers employed Naive Bayes, SVM. There has been a lot of attention paid to utilizing these capabilities to develop algorithms that can successfully study stock market trends and anticipate the stock prices thanks to the vast volumes of data made available in the current digital era and the capacity of powerful computing equipment. We begin with market data and eventually incorporate several textual data representations, ranging from straightforward representations like word counts to more complex representations involving sentiment analysis. Although regarded as unexpected, stock markets are reportedly influenced by a number of factors, including historical stock movements, hot news stories, social media sentiments, and many more.

The news headline is the summation of the news body, which necessarily lacks information yet provides important information. The decision-making process involved in choosing securities is made harder by the substantial degree of ambiguity around the nature of returns. It's important to find the correct balance between related risk and projected reward. The existing system limitations are as summarized as follows:

1)The existing system only analyses news based on sentiment for prediction, but the sentimental analysis applies to humans, it cannot differentiate between good and bad news for a particular company. Financial language can be intricate, with context-dependent jargon and nuanced meanings.

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NLP models might struggle to fully comprehend and interpret such complexity accurately.

2) It is only using news analysis for prediction, it is not using real time data from stock market for prediction.

3)Financial markets can be highly volatile, and sudden market shifts, news events, or external factors (like geopolitical events) can significantly impact stock prices. They might not always be able to predict or react to these events accurately.

IV. Proposed Methodology

This section covers the Proposed idea and the methodology of implementing a gainful stock recommendation system by integrating with reputable stock market data sources, we are able to access real-time stock market information, news, and financial reports, which allows us to recommend stocks with high profit margins.

A. Objective

The main objective of this paper is to provide two different recommendations namely for 1) Intra trading and 2) Positional Trading. Since, Intra trading is involved with trading daily, we perform Sentimental analysis on News Data. Also, We aim to classify data on the basis of Volatility to advocate correct trading methodologies. A cutting-edge model Stock Recommendation System is created to offer customers tailored stock investment advice based on their financial objectives, risk tolerance, and market research. This system makes use of machine learning algorithms, current market information, and user preferences to provide consumers with actionable stock recommendations so they can make wise investing decisions.

B. Model Architecture

The Fig.1. depicts the complete system architecture of the proposed Stock prediction and Recommendation System. The first step involves the data preprocessing of Financial News data such as Stemming , Lemmatization , Tokenization and many more. Then , the processed data is encoded and Sentimental Analysis using NLP techniques is performed on it. This results in finding the nature of Financial News collected classifying them as good and bad. On another side we use Machine learning algorithms and classify the stock data based on their Volatility. This completely preprocessed data is then split as Test and training data for Model Testing and Model training respectively. Finally, Evaluation is done to obtain the accuracy the proposed model.

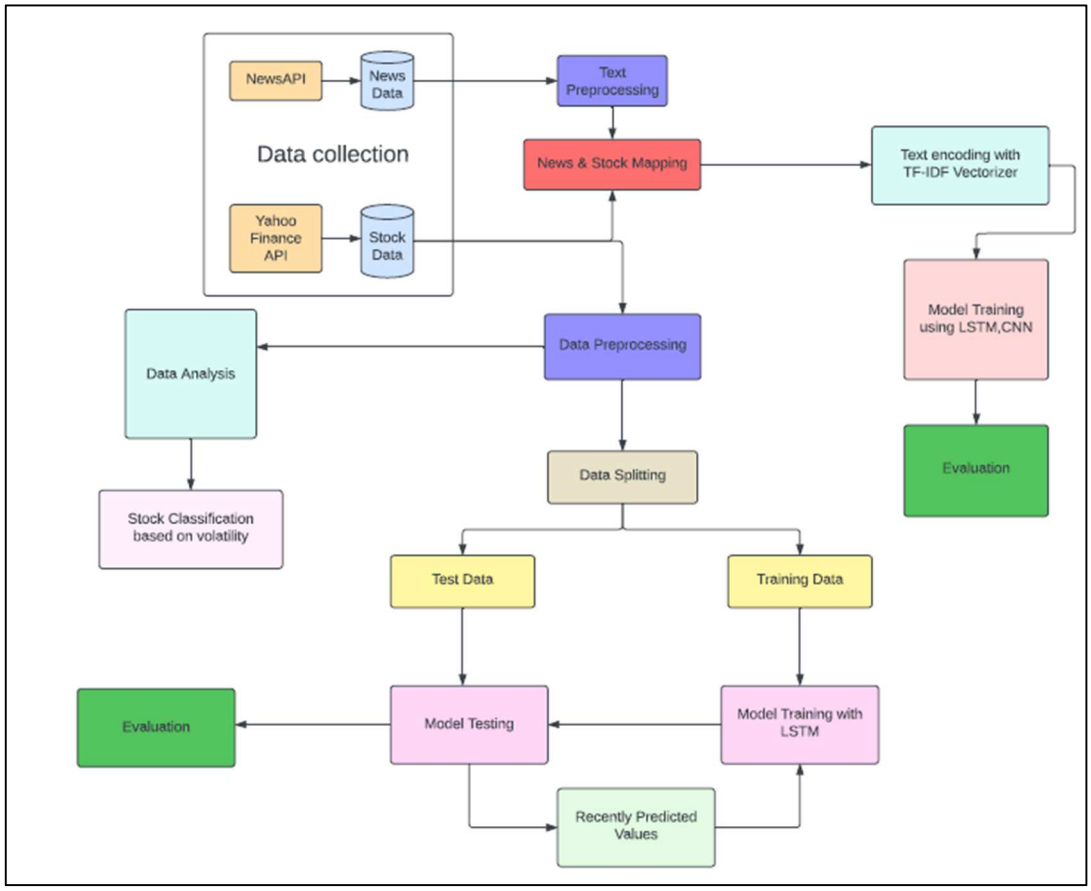

Fig.1. System Architecture

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C. Data Collection

Data for Sentimental analysis has been collected from News API, An Application Programming Interface (API) that grants access to news stories, headlines, and associated data from several news sources and publications. Stock data for both Stock Prediction and Volatility analysis has been mined from Yahoo Finance. Additionally, American Stock data has also been extracted from Alpha Vantage, which offers free stock APIs. The columns included in Stock data obtained from Alpha Vantage includes Data, Open, Close , Volume , High , Low , Adjusted close. Also the percentage change that is percentage change which is equal to the returns is mainly focused to be used in predicting the stock price. The columns included in News data mined from News APIs utilize Source, Author, title, URL and Description.

D. Data Preprocessing

To properly do sentimental analysis later, pre- processed data from the News API was obtained using the Porter-Stemmer Algorithm, WordNet Lemmatizer, and Stop word removal. Additionally, the data is vectorized using the TF-IDF vectorizer to ensure that Sentimental Analysis receives accurate input.

E. Tool Findings

The following tools have been used in our work:

1. ANACONDA:

For use in scientific and data research applications, Anaconda is an open-source version of the Python and R programming languages. It is a complete platform with more than 1,500 open-source, pre-compiled data science packages and modules for scientific computing, machine learning, and data analysis. Anaconda makes it simpler for researchers and data scientists to work with sophisticated libraries and tools.

2. TENSORFLOW:

In the fields of artificial intelligence and data science, TensorFlow is highly well-liked because it makes it easier to construct and implement machine learning models, especially deep learning models.

3. JUPYTER NOTEBOOK:

An open-source web tool called Jupyter Notebook, is used to create and share documents with live code. It is extensively utilized for interactive computing and data analysis in data science, scientific research, machine learning, and education. They offer an interactive and user- friendly environment for data exploration, data visualization, and documentation. The three primary programming languages it supports-Java, Python, and R.

F. FEATURE SELECTION

Historical Price, Volume Data and Close Price are been selected as features for Stock prediction and Volatility analysis.TF-IDF has been chosen as the suitable feature representation for Sentimental Analysis.

G. DATA ANALYSIS

The process of Volatility analysis has been carried out considering percentage change from previous day's closing price which is the percentage returns. Annualized Volatility has been used for recommending stocks on an yearly basis. Standard deviation of Percentage returns gives the historical volatility on a daily basis. Thus, we recommend Average Daily volatility should be high for intraday trading and Annualized volatility to be low for Positional Trading.

H. MODEL SELECTION

We have used LSTM models for Stock prediction as it is based on time series data.

Though LSTM models have demonstrated promise in stock price prediction, it is crucial to remember that the stock market is extremely complicated, and many elements that affect its behaviour are not captured by past price data alone. Predictions should therefore be utilized with caution, and when making investing decisions, additional pertinent facts and fundamental analysis should be taken into account. One common use of deep learning in finance is the prediction of stock prices using Long Short-Term Memory (LSTM) networks. Recurrent neural networks (RNNs), of which Long Short-Term Memory (LSTMs) are one kind, are useful for predicting stock values because they are good at identifying sequential patterns in time-series data. Sequence-related applications including time-series forecasting, speech recognition, natural language processing, and more are ideally suited for LSTMs. LSTM networks are useful for applications involving context and temporal dependencies because they can retain and learn from information received at different time steps in the past. Furthermore, a model's historical performance does not ensure future success in the stock market.

We have also used Random Forest algorithm (Ensemble based model), Logistic Regression and LSTM for Sentimental analysis. A popular machine learning algorithm for tasks in-volving regression and classification is called Random Forest. It is not a typical choice for sentiment analysis because sentiment analysis is typically approached as a text classification problem, to which other methods such as deep learning models, Naive Bayes, or Support Vector Machines (SVM) are frequently applied. With a few modifications, a RandomForest can be used for sentiment analysis, though. Every decision tree in the Random Forest bases its predictions on a set of binary choices that are decided by feature values at every node. Because decision trees can be prone to overfitting, Random Forest incorporates several trees in order to lessen this problem.

Every decision tree in the Random Forest bases its predictions on a set of binary choices that are decided by feature values at every node. Because decision trees can be prone to overfitting, Random Forest incorporates several trees in order to lessen this problem.

Applying logistic regression to sentiment analysis offers the following benefits: Because of the model's simplicity and interpretability, understanding and elucidating the predictions is made simple. When the dataset is not unduly complex and the sentiment analysis task is relatively simple, it may function well. Quick training times and computational efficiency characterize logistic regression. It is not without limitations, though: In more complex sentiment analysis tasks, it might not be able to extract

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intricate patterns from the text data as well as more sophisticated models like CNNs or LSTMs from deep learning datasets. Logistic regression's performance is highly dependent on the quality of the feature representation and may not be suitable for interpreting finer nuances in language. In actuality, the task's particular requirements and the intricacy of the data determine whether to use Logistic Regression for sentiment analysis. More sophisticated models, such as deep learning models, are frequently chosen for more difficult sentiment analysis tasks, especially in situations with a lot of text data.

V. Results

A. Experimental Setup:

To find the daily volatility of different stocks, the system first examines historical stock data. The degree of change in stock prices over time is measured by volatility. High daily volatility stocks are considered good bets for short-term intra- day trading. Handle missing values and make necessary adjustments for corporate actions (dividends, splits) to clean up the data. Complement volatility analysis with the computation of pertinent technical indicators (e.g., moving averages, relative strength index). If necessary, normalize the data.

Tokenization, stop-word removal, and text cleaning are examples of preprocessing text data. Sort every news article into three categories: neutral, negative, and positive using sentiment analysis. This usually entails calculating the daily return standard deviation over a given time period (such as the 30-day or 90-day volatility). Stocks with higher historical volatility are good bets for trading in the near term.

Press releases, social media posts, and recent news articles about the chosen stocks are gathered and examined by the system. It evaluates the sentiment of these news sources using machine learning models such as Random Forest classifiers, LSTM, and Logistic Regression. News is categorized as positive, negative, or neutral using sentiment analysis. Positively sentiment-associated stocks are thought to be better for trading decisions.

The system suggests stocks that are appropriate for intraday or short-term trading based on a combination of high daily volatility and positive sentiment. Using this data, buy and sell signals are generated with the intention of profiting from brief price movements. Stop-loss orders are among the risk management techniques that can be used to lessen possible losses.

To predict the stock prices for a given investment period which could be the short term—the system uses a stock price prediction model, such as the Long Short-Term Memory (LSTM). The model generates these forecasts based on historical stock data. Following the brief trading session, the system assesses each stock's performance. It is advised to make long-term investments in stocks that have demonstrated both the potential for long-term growth and satisfactory short- term performance. The returns attained during the brief trading period served as the foundation for these recommendations.

A. Evaluation Metrics:

Four Algorithms namely Logistic Regression, LSTM, Random Forest and CNN are used of for sentimental analysis . Accuracy for various algorithms has been calculated for various algorithms that were used for sentimental Analysis of news data and has been displayed in Table 1.From the table , it is inferred that LSTM gives the highest accuracy of 0.55 whereas Random forest gives the lowest accuracy of 0.39 .

Algorithm	Accuracy before pre-processing	Accuracy after pre-processing
Logistic Regression	0.49	0.48
LSTM	0.39	0.55
Random Forest	0.34	0.48
CNN	0.42	0.46

Accuracy = (TN + TP) / (TN + FP + TP + FN) (1)

TN – True Negative

TP – True Positive

FP – False Positive

FN– False Negative

Stock Prediction:

Table 1. Results of Sentimental Analysis

For proposed model, Mean Square Error is calculated to be evaluated. The model trained with LSTM as algorithm has given an mean square error and mean absolute error of 62.941 and 46.831487 respectively which is given in Table 2.

$$Mean\ Square\ error = \frac{\sum(Predicted - Actual)^2}{Number\ of\ Observations} \quad (2)$$

Table 2. Mean Square Error

Algorithm	Mean Square Error
LSTM	62.94127

In this section, the results of Volatility analysis and stock prediction are put on view. Fig.2 depicts the outcome of volatility analysis of "Reliance" by plotting Returns % in Y-Axis and date in X axis respectively over some time from 1990 to 2023. It is further that the volatility is significantly less but had 2 major breakthroughs. Also, the volatility is in past six months is also visibly low. Fig.3 shows the result of prediction of Stock data of today based on stock data of past 10 days. It is more accurate in a way because we use larger data of all the past 10 days. The Fig.4 shows prediction result of the finance data on the 10th day after today using today's stock market financial data. It is comparatively less accurate.

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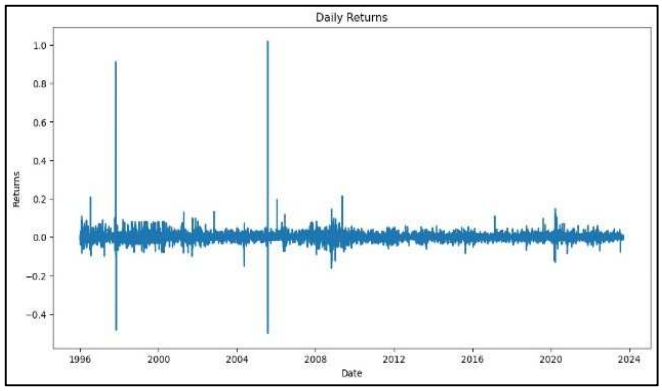

Fig.2. Volatility Analysis

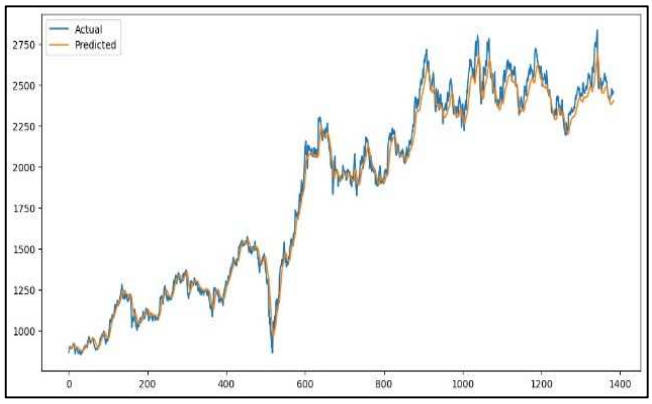

Fig.3. Prediction of stock based on past 10 days

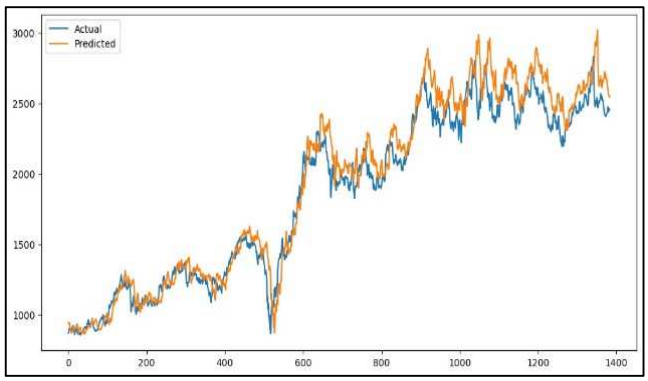

Fig.4. Prediction of Stock data after 10 days

VI. CONCLUSION

The ability of LSTM-based models to effectively capture intricate patterns and dependencies in historical stock price data has allowed the system to generate predictions and recommendations that are well-informed. The system lessens the impact of cognitive biases and human emotions, which frequently impair judgment when making financial decisions. Investors can execute trades with an advantage thanks to the system's timely recommendations, which are made possible by LSTM networks' real-time processing capabilities. But it's important to keep in mind that no stock recommendation system, even LSTM-based ones, is perfect. Numerous factors affect the financial markets, and there are inherent uncertainties in the system's predictions.

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